

MALUNG A, Sweden

WMO: 024070

Lat: 60.6765N

Lon: 13.7072E

Elev: 1004

StdP: 14.17

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-7.7	-17.6	2.2	-13.0	-11.9	3.0	-7.5	21.3	32.0	17.0	33.3	1.4	320	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	78.2	60.9	74.7	59.8	71.3	58.3	63.7	73.1	61.9	70.5	60.2	68.5	6.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.7	82.1	65.1	58.6	76.3	64.5	56.7	71.1	62.7	29.5	73.3	28.2	70.5	27.0	68.7	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	13.9	11.9	DB	-15.8	81.8	6.1	3.8	-20.2	84.5	-23.7	86.7	-27.1	88.9	-31.5	91.6
			WB	-15.5	66.0	5.6	3.0	-19.5	68.1	-22.8	69.8	-25.9	71.5	-30.0	73.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	19.6	20.9	27.7	37.5	46.5	53.9	59.3	56.2	48.5	38.7	30.1	22.2
	DBStd	16.07	11.97	11.12	9.00	6.20	6.99	5.50	5.11	5.07	5.81	7.07	8.67	12.54
	HDD50	4917	942	815	691	376	155	22	1	8	93	354	597	863
	HDD65	9683	1407	1235	1156	825	575	335	190	275	495	815	1047	1328
	CDD50	728	0	0	0	2	45	138	289	201	49	4	0	0
	CDD65	16	0	0	0	0	0	2	12	2	0	0	0	0
	CDH74	358	0	0	0	0	10	56	235	57	0	0	0	0
	CDH80	37	0	0	0	0	0	3	33	1	0	0	0	0
Wind	WSAvg	4.8	4.3	4.7	5.1	5.1	5.3	5.5	5.0	4.5	4.6	4.7	4.6	4.6
Precipitation	PrecAvg	27.7	1.7	1.3	1.3	1.7	2.1	2.9	3.5	3.0	3.0	2.7	2.5	2.0
	PrecMax	42.5	3.9	3.0	2.7	4.8	6.8	6.9	7.7	6.0	6.3	6.9	6.1	5.3
	PrecMin	19.7	0.3	0.2	0.0	0.0	0.4	0.4	1.1	0.5	0.7	0.4	0.3	0.3
	PrecStd	4.1	0.9	0.6	0.7	1.1	1.2	1.5	1.5	1.5	1.4	1.5	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	43.2	53.4	65.1	75.7	79.3	83.6	79.0	69.5	58.3	50.3	46.1
		MCWB	38.7	38.1	42.9	49.6	57.3	59.9	63.8	61.4	58.5	52.8	48.0	43.6
	2%	DB	38.7	40.5	47.1	59.6	71.3	74.8	80.1	75.3	65.0	54.5	47.6	42.5
		MCWB	36.8	37.1	39.2	46.9	55.2	58.6	62.6	61.0	55.5	50.7	45.3	40.7
	5%	DB	36.9	37.9	43.2	54.6	66.9	70.5	76.3	71.6	61.7	52.2	44.3	39.8
		MCWB	35.3	35.2	36.9	44.4	53.2	57.1	61.0	59.2	54.5	49.1	42.6	38.1
	10%	DB	34.5	35.5	40.1	50.3	61.7	66.6	72.6	67.9	58.8	49.9	41.8	37.2
		MCWB	33.3	33.3	35.4	41.9	50.5	54.9	59.4	58.0	53.1	47.3	40.4	35.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.7	40.2	43.7	51.3	59.8	63.6	66.4	64.8	60.5	54.4	48.6	44.2
		MCDB	40.6	41.9	51.3	61.6	69.9	73.1	77.2	72.3	65.5	56.6	49.7	45.8
	2%	WB	37.3	37.6	40.2	48.1	57.1	61.0	64.6	62.8	57.8	52.0	45.5	41.0
		MCDB	38.2	39.7	45.5	57.4	68.1	70.7	75.3	69.2	62.3	53.9	47.1	42.3
	5%	WB	35.4	35.6	38.0	45.4	54.4	58.5	62.8	61.2	55.9	49.8	42.9	38.2
		MCDB	36.6	37.5	42.1	53.1	64.2	68.4	73.1	68.6	60.1	51.6	44.1	39.4
	10%	WB	33.4	33.6	36.0	42.9	51.8	56.4	60.9	59.5	54.0	47.8	40.7	35.7
		MCDB	34.3	35.1	39.7	48.9	61.4	64.4	69.8	66.5	57.8	49.8	41.8	36.9
Mean Daily Temperature Range	5% DB	MDBR	12.6	15.7	17.8	18.6	20.6	19.4	20.6	19.2	17.2	13.5	10.2	12.1
		MCDBR	10.3	14.7	20.5	27.8	30.6	27.3	28.3	26.7	22.3	15.4	11.6	10.3
		MCWBR	9.6	11.9	13.5	16.4	16.2	13.6	13.7	13.7	13.8	11.9	10.8	9.7
	5% WB	MCDBR	10.1	12.5	17.9	24.8	25.6	23.3	23.8	21.5	18.1	12.1	10.6	10.2
		MCWBR	9.8	10.6	12.4	15.2	13.8	12.6	12.1	12.2	12.4	10.4	10.2	9.8
Clear-Sky Solar Irradiance	taub	0.238	0.284	0.311	0.349	0.351	0.344	0.363	0.354	0.323	0.309	0.260	N/A	
	taud	2.187	2.285	2.334	2.372	2.434	2.478	2.437	2.456	2.532	2.480	2.267	N/A	
	Ebn at Noon	169	227	259	269	277	281	272	265	256	219	158	N/A	
	Edh at Noon	12	20	27	31	32	31	32	29	23	17	11	N/A	
All-Sky Solar Radiation	RadAvg	89	297	776	1231	1495	1694	1521	1189	786	353	110	53	
	RadStd	13	56	85	145	148	114	207	123	100	67	31	6	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page